BIOL-1 Comprehensive Final Exam

College of the Redwoods, Del Norte
Instructor: Dr. Daniel Friedman

Name: ____

Date: ____

Modules 01–15 (cumulative)

Total Points: 100

Instructions: Answer **every** question in Parts **A** and **B**. In Part **C**, choose **any five** of the **seven** prompts (**5** points each). For Part **D**, follow the prompt. Budget time so Part D receives a full essay-level answer.

Part A: Multiple Choice (45 points)

Select the best answer for each question. Each question is worth 1 point. Questions 1–45 are grouped by the module they chiefly reflect.

Module 01 — Study of Life

1. Which sequence lists levels of biological organization from **smallest** to **largest** in a typical textbook hierarchy?
    - A) Atom → Molecule → Cell → Tissue → Organ
    - B) Community → Population → Organ → Cell
    - C) Organism → Population → Ecosystem → Biosphere
    - D) Ecosystem → Organ → Tissue → Cell

2. In the flow of scientific reasoning, a **hypothesis** is best described as:

    - A) A testable prediction or explanation that can be supported or refuted with evidence
    - B) A random guess that cannot be revised
    - C) The same thing as a proven theory with no further testing
    - D) A broad explanation accepted without evidence

3. **Homeostasis** refers to:

    - A) Photosynthetic capture of light energy only
    - B) Evolution of new species in one generation
    - C) Ability of an organism to maintain stable internal conditions amid external change
    - D) Random assortment of chromosomes in meiosis

Module 02 — Basic Chemistry

1. The **atomic number** of an element equals the number of:

    - A) Electrons only when ions form
    - B) Chemical bonds in DNA only
    - C) Neutrons only
    - D) Protons in its atoms (under neutral atoms, sets identity of element)

2. A **covalent bond** forms when atoms:

    - A) Avoid electrons entirely
    - B) Share electrons between nuclei
    - C) Exchange neutrons
    - D) Transfer electrons completely and stick by opposite charge alone

3. Ice floats on liquid water largely because:

    - A) Hydrogen bonding creates an open lattice in ice that is less dense than liquid water
    - B) Heat always flows out of ice faster than into water

- C) Ice is nonpolar
- D) Ice has greater density than liquid water

Module 03 — Organic Molecules

1. **Monomers** are linked into polymers such as starch or proteins by reactions that often release water; this class of reaction is called:

    - A) Hydrolysis only
    - B) Transcription
    - C) Denaturation only
    - D) Condensation / dehydration synthesis

2. Which macromolecule class typically stores **hereditary** information in cells?

    - A) Saturated fats only
    - B) Steroid hormones only
    - C) Nucleic acids (DNA / RNA)
    - D) Polysaccharide cellulose only

3. **Enzymes** are usually:

    - A) Proteins (sometimes RNA catalysts) that speed specific reactions by lowering activation energy
    - B) Inorganic salts only
    - C) DNA strands that replicate membranes
    - D) Lipids that store genetic code

Module 04 — Cells

1. The **cell theory** includes the idea that:

    - A) All cells arise only from nonliving crystal growth
    - B) Viruses satisfy all parts of the classical cell theory without exception
    - C) Only plants have cells

- D) All living organisms are made of cells and cells come from pre-existing cells (modern framing includes revisions but retains core ideas)

2. Which organelle is the primary site of **aerobic ATP harvest** (cellular respiration) in many eukaryotes?

    - A) Chloroplast
    - B) Golgi apparatus
    - C) Mitochondrion
    - D) Ribosome

3. **Prokaryotic** cells lack which structure typical of **animal** eukaryotic cells?

    - A) Membrane-bound nucleus
    - B) Cytoplasm
    - C) Ribosomes
    - D) Plasma membrane

Module 05 — Membranes

1. The **fluid mosaic** model describes the plasma membrane as:

    - A) A rigid wall of identical carbohydrate chains only
    - B) Pure DNA coated with phospholipids
    - C) A single layer of phosphates facing water on both sides only
    - D) A lipid bilayer with embedded proteins that can move laterally

2. **Simple diffusion** of a small nonpolar molecule across a lipid bilayer occurs:

    - A) Only through nuclear pores
    - B) Always against its gradient using pumps only
    - C) Down the molecule’s concentration gradient without direct metabolic energy for the crossing itself
    - D) Only with ATP hydrolysis each time

3. **Osmosis** refers specifically to:

    ◦ A) Active transport of proteins into mitochondria only
    ◦ B) Diffusion of **water** across a selectively permeable membrane
    ◦ C) Movement of solute against its gradient with ATP
    ◦ D) Random mutation of membrane lipids

Module 06 — Metabolism

1. **ATP** is useful to cells mainly because:

    ◦ A) It is only made in the nucleus
    ◦ B) It is permanent storage that never releases phosphate
    ◦ C) It replaces DNA during replication
    ◦ D) Energy stored in its phosphate bonds can be transferred to drive cellular work when hydrolyzed / regenerated in coupled reactions

2. In overview, **photosynthesis** builds sugars using light energy; **cellular respiration**:

    ◦ A) Creates glucose from carbon dioxide without energy input
    ◦ B) Uses oxygen and breaks fuel molecules to capture usable energy (often linked to ATP production)
    ◦ C) Copies RNA using ribosomes only
    ◦ D) Occurs only in plant roots at night only

3. An **enzyme's active site**:

    ◦ A) Binds substrates and catalyzes conversion to products for that reaction
    ◦ B) Destroys all substrates permanently without reuse
    ◦ C) Is identical on every enzyme regardless of shape
    ◦ D) Stores sunlight like chlorophyll exclusively

Module 07 — Molecular Genetics

1. The central dogma in many cells reads **DNA → RNA → protein**. **Translation** refers to:

    ◦ A) Building RNA from a DNA template

- ◦ B) Ribosomes synthesizing polypeptides using an mRNA code
- ◦ C) Methylation of histones only
- ◦ D) DNA replication only

2. During **DNA replication**, new strands are elongated primarily by:

   - ◦ A) Lipase breaking fats
   - ◦ B) ATP synthase in thylakoids only
   - ◦ C) DNA polymerase adding complementary nucleotides
   - ◦ D) RNA polymerase reading ribosomes

3. **RNA** differs from DNA in a typical introductory biology contrast because RNA:

   - ◦ A) Contains ribose (often single-stranded functional molecules such as mRNA, tRNA, rRNA)
   - ◦ B) Is usually double-stranded helical archive in cytoplasm only
   - ◦ C) Uses thymine instead of uracil always
   - ◦ D) Cannot leave the nucleus ever

Module 08 — Cellular Genetics

1. A human **somatic** cell is diploid (2n); a mature **gamete** is:

   - ◦ A) Haploid (n)
   - ◦ B) Triploid always
   - ◦ C) 4n after meiosis II always
   - ◦ D) Diploid (2n)

2. **Crossing over** between homologous chromosomes primarily occurs during:

   - ◦ A) DNA replication in G2 only
   - ◦ B) Cytokinesis in bacteria only
   - ◦ C) Metaphase of mitosis only
   - ◦ D) Prophase I of meiosis

3. **Mitosis** followed by cytokinesis of one diploid parent cell typically yields:

    ◦ A) Four genetically different haploid nuclei
    ◦ B) One haploid and one diploid cell always
    ◦ C) Two genetically identical diploid daughter cells (barring mutation)
    ◦ D) Eight cells from one division

Module 09 — Inheritance Genetics

1. Alternative versions of one gene (such as flower-color alleles) are called:

    ◦ A) Phenotypes only
    ◦ B) Chromatids
    ◦ C) Alleles
    ◦ D) Barr bodies

2. For complete dominance, cross **Aa** × **Aa**. Expected **phenotypic** ratio among offspring is often:

    ◦ A) All heterozygotes identical to both parents always
    ◦ B) **3 dominant-looking : 1 recessive-looking** among offspring (classic single-gene expectation)
    ◦ C) 9 : 3 : 3 : 1 for one gene
    ◦ D) 1 : 1 only

3. **Incomplete dominance** often yields offspring ratios such as **1 : 2 : 1** phenotypes when two heterozygotes mate because:

    ◦ A) Gene flow removes heterozygotes every generation
    ◦ B) Dominant allele silences every heterozygote completely
    ◦ C) Crossing over erases alleles
    ◦ D) Heterozygotes show an intermediate phenotype between homozygotes

Module 10 — Epigenetics

1. **Epigenetics** emphasizes changes in trait expression that:
    - A) Occur only in prokaryotes
    - B) Always change DNA nucleotide order immediately and irreversibly only one way
    - C) Alter gene activity **without** necessarily changing the DNA sequence itself
    - D) Replace mitosis entirely
2. DNA **methylation** associated with gene silencing fits the memory hook "methylation ≈ muting" because methylation often:
    - A) Converts RNA into protein without ribosomes
    - B) Forces frameshift mutations automatically
    - C) Reduces transcription / promotes a gene-off state compared with unmethylated promoter regions in many examples
    - D) Doubles chromosome number instantly
3. In female mammals, **X-inactivation** produces a dense **Barr body** because:
    - A) Both X chromosomes replicate only in mitochondria
    - B) One X is largely condensed / silenced in each somatic cell for dosage compensation
    - C) Every X is deleted from the genome
    - D) Meiosis completes in liver cells daily

Module 11 — Genomics & Biotechnology

1. **PCR** is used mainly to:
    - A) Destroy RNA selectively only
    - B) Fuse gametes in vitro only
    - C) Replace photosynthesis in chloroplasts
    - D) Amplify targeted DNA sequences from small samples

2. In standard **agarose gel electrophoresis**, DNA fragments migrate toward:
    - A) The positive electrode; smaller fragments often travel farther in a given time
    - B) Only RNA, never DNA
    - C) The middle band regardless of size always
    - D) The negative electrode because DNA is positively charged
3. **CRISPR-Cas9** is widely described as a tool that can:
    - A) Target specific DNA sequences for cutting / editing with guide RNA assistance
    - B) Increase mutation rates only in mitochondria randomly without targeting
    - C) Eliminate all epigenetic marks globally without sequence specificity
    - D) Replace osmosis in plant roots

Module 12 — Darwin & Evolution

1. **Natural selection** requires heritable variation and differential reproductive success linked to traits in a given environment; which statement adds Darwin's observation about numbers of offspring?
    - A) Every individual leaves equal offspring regardless of traits
    - B) Evolution cannot change allele frequencies
    - C) Populations tend to produce **more offspring than can survive and reproduce**
    - D) Acquired traits always become alleles immediately
2. **Homologous** tetrapod limbs with different functions suggest:
    - A) No shared ancestry possible
    - B) Shared ancestry with structural modification under different selective regimes
    - C) That analogous structures cannot exist
    - D) Identical DNA sequences in every species without difference
3. **Evolutionary fitness** in population genetics usage emphasizes:
    - A) Relative contribution of alleles to future generations through survival and reproduction
    - B) Survival years without offspring equally weighted as reproduction

- ◦ C) Ability to learn calculus
- ◦ D) Bench press strength across species

Module 13 — How Populations Evolve

1. **Gene flow** tends to:

    - ◦ A) Occur only during bottlenecks always upward
    - ◦ B) Move alleles between populations through migration and mating—often homogenizing allele frequencies between connected groups over time (depending on rates)
    - ◦ C) Replace mutation entirely
    - ◦ D) Reduce allele sharing between populations always to zero instantly

2. **Genetic drift** has **strongest** effects when:

    - ◦ A) Population size is **small** so random sampling shifts allele frequencies strongly between generations
    - ◦ B) Population size is very large and migration is zero
    - ◦ C) Hardy–Weinberg assumptions all hold perfectly forever
    - ◦ D) Selection coefficients are exactly zero in infinite populations without variation

3. The Hardy–Weinberg equilibrium serves chiefly as:

    - ◦ A) A substitute definition for biological species
    - ◦ B) Proof evolution never happens on islands
    - ◦ C) Measurement of carrying capacity only
    - ◦ D) A mathematical **null / baseline** against which observed allele-frequency change is interpreted

Module 14 — Macroevolution

1. Different frog species breed in different seasons, preventing hybridization. This is a **prezygotic** barrier classified as:

    - ◦ A) Hybrid breakdown after many generations only

- ◦ B) Temporal isolation
- ◦ C) Reduced hybrid fertility only
- ◦ D) Bottleneck effect

2. Sterile **mule** hybrids illustrate:

   - ◦ A) Sympatric polyploidy in mammals always
   - ◦ B) Prezygotic mechanical isolation only
   - ◦ C) Directional selection on horses only every generation
   - ◦ D) Postzygotic reduced hybrid fertility

3. A mountain range splits one fish species into non-interbreeding lineages—classic mode:

   - ◦ A) Population crashes without allele-frequency change
   - ◦ B) Sympatric speciation via instant mutation only
   - ◦ C) Only artificial selection
   - ◦ D) Allopatric speciation starting with geographic isolation

Module 15 — Population & Systems Ecology

1. A population rising steeply when resources are abundant shows **J-shaped exponential** growth until limits bite; **logistic** growth differs mainly because:

   - ◦ A) Carrying capacity is always zero
   - ◦ B) Growth slows as **N** approaches **K** (carrying capacity)
   - ◦ C) Growth rate never changes from zero
   - ◦ D) Birth rate equals death rate only when individuals stop eating

2. A disease spreading faster when hosts are crowded is typically treated as:

   - ◦ A) Proof energy cycles identically to nitrogen atoms
   - ◦ B) Only abiotic temperature effects
   - ◦ C) Density-dependent regulation (impact intensifies with density in many textbook examples)
   - ◦ D) Density-independent always because pathogens are small

3. In food webs, chemical nutrients cycle while energy from sunlight through producers is ultimately:

   ◦ A) Created anew by decomposers from vacuum energy each step
   ◦ B) Lost largely as heat between trophic transfers—supporting fewer top consumers than low trophic levels per unit primary production
   ◦ C) Recycled photon-for-photon inside lions forever without loss
   ◦ D) Stored 100% at apex predators only

Part B: Fill in the Blank (15 points)

Questions 1–15. Each answer is worth 1 point. Use the single best term from the word bank (exact wording and capitalization as listed—the key notes accepted minor hyphen/spacing variants only where stated).

Word bank: Alleles, Bottleneck, Cell wall, Codominance, Decomposers, Denaturation, Endocytosis, Epigenetics, Exponential, Founder, Golgi apparatus, **Hardy–Weinberg**, Heterozygous, Homeostasis, Hydrogen, Lysosome, Mitochondria, Osmosis, Translation

(Bank lists 19 terms; four are not used as correct answers.)

1. Maintaining stable internal conditions despite external fluctuations is called __________________.

2. Water molecules stick to each other through __________________ bonding (partial charges on O–H).

3. Cells take in large particles by vesicle wrapping using processes such as phagocytosis, a form of __________________.

4. Membrane-bound sac of hydrolytic enzymes that degrades worn cell parts in many animal cells: __________________.

5. Organelles that modify, sort, and ship many proteins are stacks of the __________________.

6. High heat or harsh pH often unfolds a protein's three-dimensional shape (without necessarily breaking every peptide bond); this loss of native structure is called __________________.

7. Movement of **water** across a selectively permeable membrane, down its concentration gradient, is called __________________.

8. In many eukaryotic cells, **aerobic ATP harvest** is concentrated in organelles called __________________.

9. An organism with genotype **Aa** for one gene is __________________ at that locus.

10. Type **AB** blood with both **A** and **B** alleles expressed illustrates __________________.

11. DNA methylation that silences promoters without changing letter sequence is discussed under __________________ along with histone changes.

12. After a random catastrophe removes most individuals, surviving allele frequencies may shift by drift through a __________________ effect.

13. When a few colonists seed an island population, allele frequencies may reflect only colonists' genes—a __________________ effect.

14. The equilibrium (p^2 + 2pq + q^2 = 1) under ideal assumptions is tied to the __________________ population-genetics baseline model.

15. Fungi and bacteria returning nutrient elements from dead matter toward producers are major __________________ in ecosystems.

Part C: Integrated Short Answer (25 points)

Choose any five of the seven numbered prompts below. 5 points each. Label each answer with its prompt number. Write clearly in complete sentences in the space provided.

1. Compare **mitosis** and **meiosis** for a diploid animal cell: number of divisions, number of nuclear products, **ploidy** of products, genetic similarity among products, and **one** biological purpose for each process.

Answer:

2. Trace **information flow** from a gene to a functional protein (major stages). Add **one** sentence on how **epigenetic** marking could reduce protein output **without** changing the DNA sequence.

Answer:

3. Explain **natural selection** using **variation**, **overproduction**, and **differential reproductive success**. Then distinguish **analogous** versus **homologous** structures in **two** sentences.

Answer:

4. Contrast **genetic drift** and **natural selection** as mechanisms changing allele frequencies. Give **one** example each of **bottleneck** and **founder** effects in plain language.

Answer:

5. Compare **exponential** versus **logistic** population growth and define **carrying capacity (K)**. Give **one density-dependent** and **one density-independent** limiting factor (full sentences).

Answer:

6. Compare **photosynthesis** and **cellular respiration** at an overview level: **what** each process broadly accomplishes for the cell, **where** much of each pathway occurs in many eukaryotic cells, and **one** sentence on how the two pathways relate as complementary parts of energy and carbon flow.

Answer:

7. State the **biological species concept** at an introductory level. Then give **one** example of a **prezygotic** barrier and **one** example of a **postzygotic** barrier (full sentences; names of barrier categories are fine).

Answer:

Part D: Essay (15 points)

Choose ONE essay prompt. Write a coherent answer several paragraphs long with an introduction and conclusion. Use accurate vocabulary from multiple units.

Option A — From molecules to populations. Explain how **DNA**, **genes**, and **alleles** relate to **traits** in populations. Connect **mutation**, **natural selection**, **genetic drift**, and **gene flow** to **allele-frequency change**. Finish by explaining why **populations** evolve rather than single individuals.

Option B — Energy, matter, and humans. Describe **energy flow** versus **nutrient cycling** using **producers**, **consumers**, and **decomposers**. Apply the **~10% rule** to explain why eating **lower** on the food chain generally puts **less** demand on **primary production** than feeding the same number of people largely on **apex carnivores**.

Option C — Speciation and isolation. Summarize the **biological species concept** at an introductory level. Compare **prezygotic** and **postzygotic** barriers with **two** examples total (one of each type). Explain **allopatric** versus **sympatric** speciation and give **one** plant-focused example for sympatric routes.

Essay Option Letter (A, B, or C): _____

Feedback (Optional)

Is there anything your instructor should know about how you approached this final?

End of Comprehensive Final Exam